

analysis

Matilde 2026

2026-08-22

```
#install packages (see ../requirements.txt)
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2     4.0.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(tidyr)
library(rvest)
```

```
##
## Attaching package: 'rvest'
##
## The following object is masked from 'package:readr':
##
##     guess_encoding
```

```
#setting directories:
# input data (the analytical sample)
analysis_path <- file.path("../", "data", "analysis_data")
# create directory if it does not exist
if (!dir.exists(analysis_path)) {
  dir.create(analysis_path)
}

out_dir <- file.path("../", "out")
dir.create(
  out_dir,
  recursive = TRUE,
  showWarnings = FALSE
)

# setting figure output directory
figures_dir <- file.path(out_dir, "figures")
dir.create(
  figures_dir,
  recursive = TRUE,
  showWarnings = FALSE
)
```

```
# analytical sample:
df <- read_csv(file.path(analysis_path, "analysis_df.csv"))
```

```
## Rows: 3977 Columns: 36
## — Column specification —————
## Delimiter: ","
## chr  (24): sex_series, transcription_url, archive_image_api, name, sex, occu...
## dbl  (10): series_page, crowd_id, archive_image_id, patient_id, age_at_admis...
## date  (2): date_of_birth, admission_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# checking dataframe structure and dimensions
str(df) #should be comma-separated (3997 rows, 36 columns)
```

```

## spc_tbl_ [3,977 × 36] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ sex_series          : chr [1:3977] "women" "women" "women" "women" ...
## $ series_page         : num [1:3977] 3 3 3 3 3 3 3 3 3 3 ...
## $ crowd_id            : num [1:3977] 1532962 1532962 1532962 1532962 1532962
...
## $ archive_image_id    : num [1:3977] 63211977 63211977 63211977 63211977 63211
977 ...
## $ transcription_url    : chr [1:3977] "https://cs.rigsarkivet.dk/picture/view-o
rigin-values/63211977" "https://cs.rigsarkivet.dk/picture/view-origin-values/6321197
7" "https://cs.rigsarkivet.dk/picture/view-origin-values/63211977" "https://cs.rigsar
kivet.dk/picture/view-origin-values/63211977" ...
## $ archive_image_api    : chr [1:3977] "https://api.rigsarkivet.dk/ao/v1/images/
63211977" "https://api.rigsarkivet.dk/ao/v1/images/63211977" "https://api.rigsarkive
t.dk/ao/v1/images/63211977" "https://api.rigsarkivet.dk/ao/v1/images/63211977" ...
## $ patient_id          : num [1:3977] 4609 4611 4613 4618 4619 ...
## $ name                 : chr [1:3977] "Marenline Johanne Plassmann" "Larsine Ni
elsen" "Dorthe Marie Johansen Pind (f. Boye)" "Severinel Dorthea Sørensen (Hvolris)"
...
## $ sex                  : chr [1:3977] "k" "k" "k" "k" ...
## $ occupation           : chr [1:3977] "Glaspusters Enke" "Fiskehandlers Hustru"
"Husmands Hustru" "Tjenestepige" ...
## $ residence            : chr [1:3977] "Aalborg" "Frederikshavn" "Kolind Bro S.
Sønder H. Randers A." "Hvolris, Bjerregrav S. Rinds H. Viborg A." ...
## $ age_at_admission     : num [1:3977] 40 32 35 48 36 46 36 60 30 45 ...
## $ date_of_birth        : Date[1:3977], format: NA NA ...
## $ religion             : chr [1:3977] "EL" "EL" "EL" "EL" ...
## $ marital_status       : chr [1:3977] "enke" "gift" "gift" "ugift" ...
## $ admission_date       : Date[1:3977], format: "1889-09-24" "1889-09-26" ...
## $ admitted_from       : chr [1:3977] "St. Hans Hospital" "Frederikshavn Sygehu
s" "Hjemmet" "Hobro Sygehus" ...
## $ diagnosis_at_admission : chr [1:3977] "Melankoli" "Mani." "Melankoli" "Forrykth
ed" ...
## $ duration_at_admission : chr [1:3977] "2 Aar" "1-2 Maan." "ubekj." "ubekj." ...
## $ journal_number       : chr [1:3977] "169-89" "193-89" "198-89" "209-89" ...
## $ care_payment_class   : chr [1:3977] "3 1/2 bet." "3 1/2 bet." "3 1/2 bet."
"3" ...
## $ episode_number       : chr [1:3977] "1" "1" "c. 5." "2" ...
## $ remarks              : chr [1:3977] NA NA NA NA ...
## $ family_disposition   : chr [1:3977] "ja" NA "ja" NA ...
## $ discharge_number      : num [1:3977] 4341 4120 4123 4915 4116 ...
## $ discharge_date       : chr [1:3977] "1891-04-21" "1889-12-15" "1889-12-18" "1
895-02-06" ...
## $ condition_at_discharge : chr [1:3977] "helbredet" "helbredet" "helbredet" "død"
...
## $ discharged_to        : chr [1:3977] "Til Kjöbenhavn" "Til Hjemmet" "Til Hjemm
et" NA ...
## $ previous_number      : chr [1:3977] NA NA NA NA ...
## $ readmitted_under_number: chr [1:3977] NA NA NA NA ...
## $ transcriber_notes     : chr [1:3977] NA NA NA NA ...
## $ record_key           : num [1:3977] 1 2 3 4 5 6 7 8 9 10 ...
## $ patient_key          : chr [1:3977] "women_4609" "women_4611" "women_4613" "w
omen_4618" ...
## $ birth_year           : num [1:3977] NA NA NA NA NA NA NA NA NA ...
## $ admission_year       : num [1:3977] 1889 1889 1889 1889 1889 ...
## $ discharge_year       : num [1:3977] 1891 1889 1889 1895 1889 ...

```

```
## - attr(*, "spec")=
## .. cols(
## ..   sex_series = col_character(),
## ..   series_page = col_double(),
## ..   crowd_id = col_double(),
## ..   archive_image_id = col_double(),
## ..   transcription_url = col_character(),
## ..   archive_image_api = col_character(),
## ..   patient_id = col_double(),
## ..   name = col_character(),
## ..   sex = col_character(),
## ..   occupation = col_character(),
## ..   residence = col_character(),
## ..   age_at_admission = col_double(),
## ..   date_of_birth = col_date(format = ""),
## ..   religion = col_character(),
## ..   marital_status = col_character(),
## ..   admission_date = col_date(format = ""),
## ..   admitted_from = col_character(),
## ..   diagnosis_at_admission = col_character(),
## ..   duration_at_admission = col_character(),
## ..   journal_number = col_character(),
## ..   care_payment_class = col_character(),
## ..   episode_number = col_character(),
## ..   remarks = col_character(),
## ..   family_disposition = col_character(),
## ..   discharge_number = col_double(),
## ..   discharge_date = col_character(),
## ..   condition_at_discharge = col_character(),
## ..   discharged_to = col_character(),
## ..   previous_number = col_character(),
## ..   readmitted_under_number = col_character(),
## ..   transcriber_notes = col_character(),
## ..   record_key = col_double(),
## ..   patient_key = col_character(),
## ..   birth_year = col_double(),
## ..   admission_year = col_double(),
## ..   discharge_year = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
#for analysis, transcription notes are omitted, and treated as NA:
```

```
df <- df %>%  
  mutate(  
    across(  
      where(is.character),  
      ~ na_if(.x, "could_not_transcribe")  
    )  
  )
```

```
# changing data types:
```

```
df <- df %>%  
  mutate(  
    # Standardizing dates  
    admission_date = as.Date(admission_date),  
    discharge_date = as.Date(discharge_date),  
    date_of_birth = as.Date(date_of_birth),  
  
    # Extracting years  
    birth_year = lubridate::year(date_of_birth),  
    admission_year = lubridate::year(admission_date),  
    discharge_year = lubridate::year(discharge_date),  
  
    # Factorizing defined categorical variables  
    sex_series = factor(sex_series),  
    sex = factor(sex),  
    marital_status = factor(marital_status),  
  
    # Numeric / sequential variables  
    series_page = as.integer(series_page),  
    crowd_id = as.integer(crowd_id),  
  
    #cat:  
    patient_key = as.character(patient_key)  
  )
```

DESCRIPTIVE ANALYSES

```
# Diagnoses columns  
#what diagnostic language will I find? (not standardized)  
diagnosis_counts <- df %>%  
  count(diagnosis_at_admission, sort = TRUE)  
diagnosis_counts
```

```
## # A tibble: 725 × 2
##   diagnosis_at_admission      n
##   <chr>                    <int>
## 1 Melankoli                926
## 2 Mani                     813
## 3 Sløvsind                 346
## 4 Forrykthed               293
## 5 Dem. præc.               89
## 6 Melank.                  80
## 7 Vanvid                   52
## 8 Fatuitet                 45
## 9 Mel.                    45
## 10 Dem. præc              41
## # i 715 more rows
```

```
# condition columns
# how were conditions structured?
condition_count <- df %>%
  count(condition_at_discharge, sort = TRUE)
condition_count
```

```
## # A tibble: 217 × 2
##   condition_at_discharge      n
##   <chr>                    <int>
## 1 død                     885
## 2 bedret                 710
## 3 helbr                  590
## 4 uhelbr                 498
## 5 helbredet              427
## 6 <NA>                   265
## 7 uhelbredet             171
## 8 helbredt                69
## 9 ikke sindssyg          67
## 10 uhelbredt              31
## # i 207 more rows
```

```
cat("Unique discharge conditions:",nrow(condition_count))
```

```
## Unique discharge conditions: 217
```

```
# bedret, helbredet, uhelbredet/uforandret, død, på prøve)
```

```

# Recoding obvious textual variants
diagnosis_recode <- c(
  "melank" = "melankoli",
  "mel"    = "melankoli",
  "melancholi" = "melankoli",
  "melanch" = "melankoli",
  "mekankoli" = "melankoli",
  "melankol" = "melankoli",
  "forv" = "forvirring",
  "dements" = "demens",
  "dementia" = "demens",
  "mani period" = "mani periodisk",
  "firrykthed" = "forrykthed",
  "forrkt" = "forrykthed",
  "forrykhed" = "forrykthed",
  "forrykt" = "forrykthed",
  "dem prac" = "dementia præcox",
  "dem præcox" = "dementia præcox",
  "dem præc" = "dementia præcox",
  "dem pre" = "dementia præcox"
)

# Standardizing diagnosis strings for analysis
df <- df %>%
  mutate(
    diagnosis_standardized = diagnosis_at_admission %>%
      str_to_lower() %>%
      str_remove_all("[[:punct:]]") %>%
      str_squish() %>%
      na_if(""),

    diagnosis_standardized = case_when(
      str_detect(diagnosis_standardized, "obs") ~ "obs",
      TRUE ~ diagnosis_standardized
    ),

    diagnosis_standardized = recode(
      diagnosis_standardized,
      !!!diagnosis_recode,
      .default = diagnosis_standardized
    )
  )
df <- df %>%
  mutate(
    diagnosis_standardized = recode(
      diagnosis_standardized,
      !!!diagnosis_recode,
      .default = diagnosis_standardized
    )
  )

# viewing the standardized diagnoses
clean_diagnosis_counts <- df %>%
  count(diagnosis_standardized, sort = TRUE)
clean_diagnosis_counts

```

```
## # A tibble: 496 × 2
##   diagnosis_standardized    n
##   <chr>                  <int>
## 1 melankoli              1117
## 2 mani                   839
## 3 sløvsind               351
## 4 forrykthed             311
## 5 dementia præcox        241
## 6 obs                    76
## 7 vanvid                 56
## 8 demens                 52
## 9 fatuitet               47
## 10 forvirring            44
## # i 486 more rows
```

```
cat("Number of clean diagnosis:", nrow(clean_diagnosis_counts)) #495
```

```
## Number of clean diagnosis: 496
```

```
# Comparing diagnoses across the two archival series
```

```
diagnosis_by_series <- df %>%
  count(
    sex_series,
    diagnosis_standardized
  ) %>%
  group_by(sex_series) %>%
  mutate(
    proportion = n / sum(n)
  ) %>%
  arrange(sex_series, desc(n)) %>%
  ungroup()
```

```
diagnosis_by_series
```

```
## # A tibble: 554 × 4
##   sex_series diagnosis_standardized    n proportion
##   <fct>      <chr>                  <int>      <dbl>
## 1 men      melankoli              419    0.209
## 2 men      mani                   362    0.181
## 3 men      sløvsind               219    0.109
## 4 men      forrykthed             162    0.0808
## 5 men      dementia præcox        135    0.0673
## 6 men      obs                    64    0.0319
## 7 men      fatuitet                37    0.0185
## 8 men      vanvid                 33    0.0165
## 9 men      demens                 32    0.0160
## 10 men     sløvsind d p           17    0.00848
## # i 544 more rows
```



```
# Top 10 most frequent standardized diagnoses overall
top_10_diagnoses <- df %>%
  filter(!is.na(diagnosis_standardized)) %>%
  count(diagnosis_standardized, sort = TRUE) %>%
  slice_head(n = 10) %>%
  pull(diagnosis_standardized)

# Overall: top 10 diagnoses over time
diagnosis_over_time <- df %>%
  filter(
    !is.na(diagnosis_standardized),
    !is.na(admission_year)
  ) %>%
  group_by(admission_year) %>%
  mutate(total_diagnoses = n()) %>%
  ungroup() %>%
  filter(diagnosis_standardized %in% top_10_diagnoses) %>%
  count(
    admission_year,
    diagnosis_standardized,
    total_diagnoses
  ) %>%
  mutate(
    proportion = n / total_diagnoses,
    diagnosis_standardized = factor(
      diagnosis_standardized,
      levels = top_10_diagnoses
    )
  )

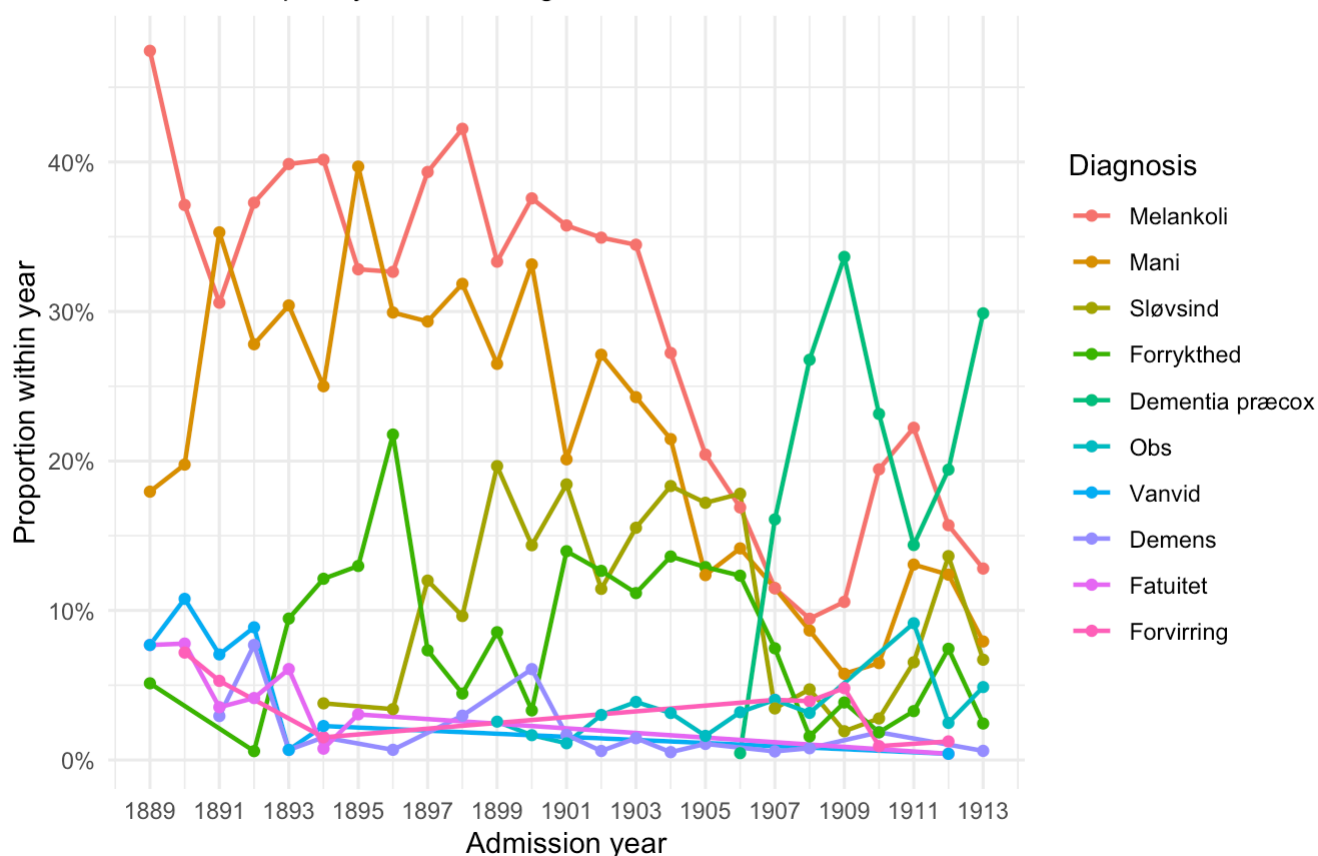
# By sex: same top 10 diagnoses over time
diagnosis_over_time_sex <- df %>%
  filter(
    !is.na(diagnosis_standardized),
    !is.na(admission_year)
  ) %>%
  group_by(sex_series, admission_year) %>%
  mutate(total_diagnoses = n()) %>%
  ungroup() %>%
  filter(diagnosis_standardized %in% top_10_diagnoses) %>%
  count(
    sex_series,
    admission_year,
    diagnosis_standardized,
    total_diagnoses
  ) %>%
  mutate(
    proportion = n / total_diagnoses,
    diagnosis_standardized = factor(
      diagnosis_standardized,
      levels = top_10_diagnoses
    )
  )
```

```
# Plotting overall distribution of diagnoses over time
diagnosis_over_time_plot <- ggplot(
  diagnosis_over_time,
  aes(
    x = admission_year,
    y = proportion,
    color = diagnosis_standardized
  )
) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_x_continuous(
    breaks = seq(1889, 1914, by = 2)
  ) +
  scale_y_continuous(labels = scales::percent) +
  scale_color_discrete(
    labels = \(x) str_to_sentence(x)
  ) +
  labs(
    title = "Most Frequent Diagnoses Over Time",
    subtitle = "Ten most frequently recorded diagnoses overall",
    x = "Admission year",
    y = "Proportion within year",
    color = "Diagnosis"
  ) +
  theme_minimal()
```

diagnosis_over_time_plot

Most Frequent Diagnoses Over Time

Ten most frequently recorded diagnoses overall

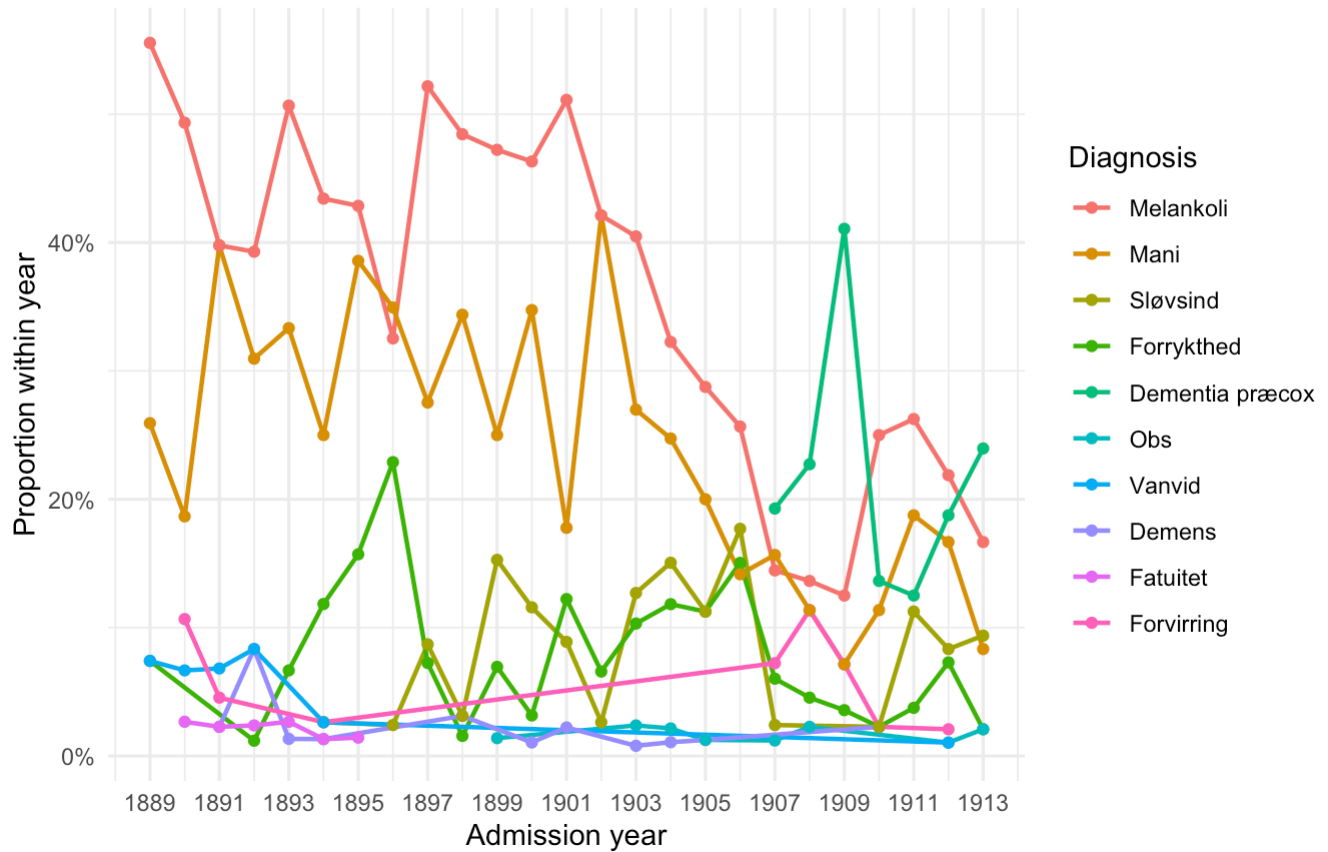


```
# same plot, but for women:
diagnosis_over_time_women_plot <- diagnosis_over_time_sex %>%
  filter(sex_series == "women") %>%
  ggplot(
    aes(
      x = admission_year,
      y = proportion,
      color = diagnosis_standardized
    )
  ) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_x_continuous(
    breaks = seq(1889, 1914, by = 2)
  ) +
  scale_y_continuous(labels = scales::percent) +
  scale_color_discrete(
    labels = \(x) str_to_sentence(x)
  ) +
  labs(
    title = "Most Frequent Diagnoses Over Time: Women",
    subtitle = "Ten most frequently recorded diagnoses overall",
    x = "Admission year",
    y = "Proportion within year",
    color = "Diagnosis"
  ) +
  theme_minimal()

diagnosis_over_time_women_plot
```

Most Frequent Diagnoses Over Time: Women

Ten most frequently recorded diagnoses overall



```
sort(unique(df$admission_year))
```

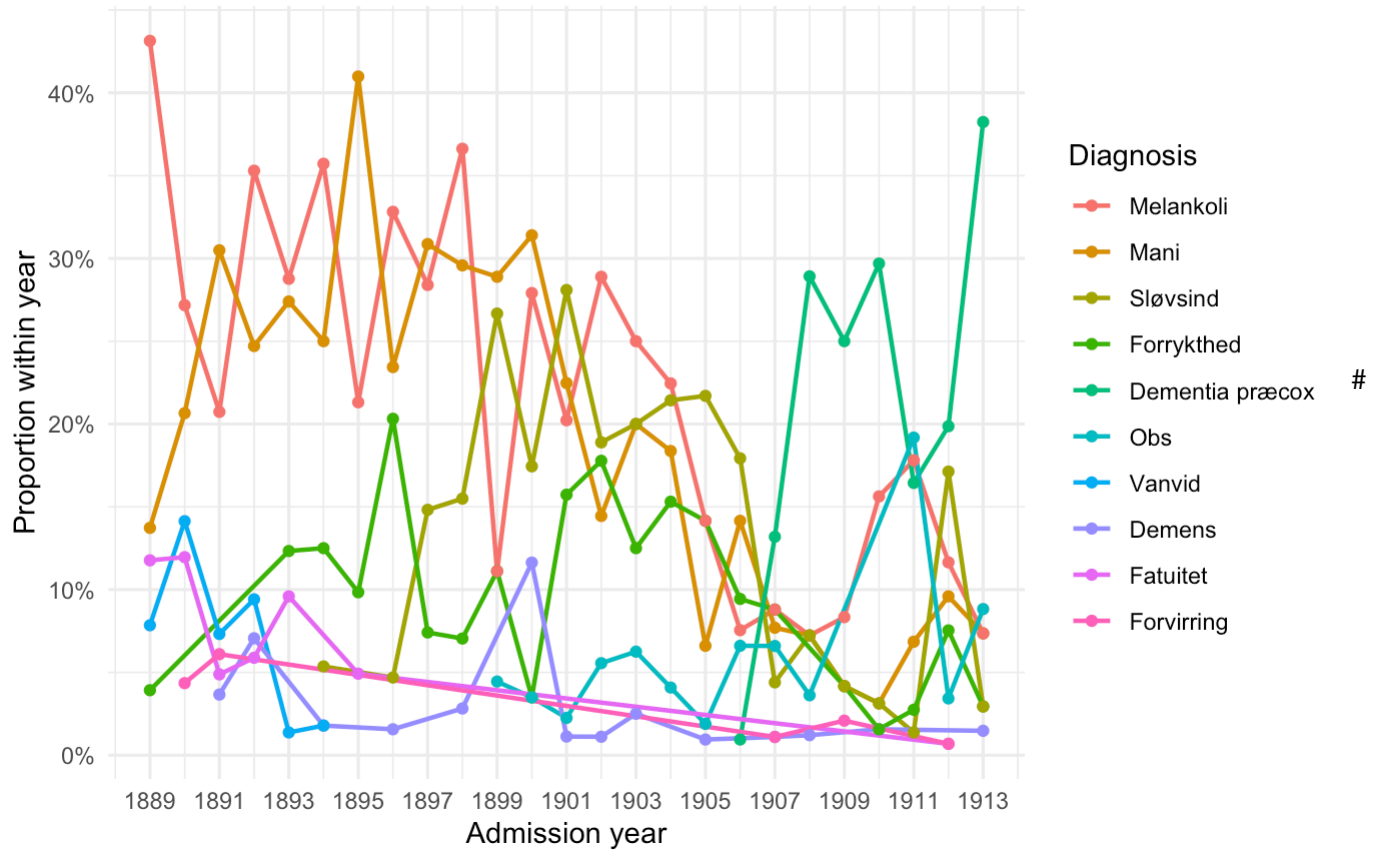
```
## [1] 1889 1890 1891 1892 1893 1894 1895 1896 1897 1898 1899 1900 1901 1902 1903
## [16] 1904 1905 1906 1907 1908 1909 1910 1911 1912 1913
```

```
# Same plot, but for men
# Men
diagnosis_over_time_men_plot <- diagnosis_over_time_sex %>%
  filter(sex_series == "men") %>%
  ggplot(
    aes(
      x = admission_year,
      y = proportion,
      color = diagnosis_standardized
    )
  ) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_x_continuous(
    breaks = seq(1889, 1914, by = 2)
  ) +
  scale_y_continuous(labels = scales::percent) +
  scale_color_discrete(
    labels = \(x) str_to_sentence(x)
  ) +
  labs(
    title = "Most Frequent Diagnoses Over Time: Men",
    subtitle = "Ten most frequently recorded diagnoses overall",
    x = "Admission year",
    y = "Proportion within year",
    color = "Diagnosis"
  ) +
  theme_minimal()

diagnosis_over_time_men_plot
```

Most Frequent Diagnoses Over Time: Men

Ten most frequently recorded diagnoses overall



```

# from inspection of unique condition string:
condition_recode <- c(
  "helbredet" = "helbredt",
  "helbr"     = "helbredt",
  "uhelbredet" = "uhelbredt",
  "uforandr"  = "uhelbredt",
  "uforand"   = "uhelbredt",
  "uforandret" = "uhelbredt",
  "sindsyg"   = "uhelbredt",
  "genindlagt" = "genindlagt",
  "uhelbr"    = "uhelbredt",
  "uhelbredelig" = "uhelbredt",
  "uh"        = "uhelbredt",
  "uhel"      = "uhelbredt",
  "uhelbre"   = "uhelbredt",
  "bedr"      = "bedret",
  "bedring"   = "bedret",
  "bedrt"     = "bedret",
  "bedring"   = "bedret",
  "bredret"   = "bedret",
  "paa prøve" = "på prøve",
  "afdød"     = "død",
  "ikke sindsyg" = "ikke sindssyg"
)

df <- df %>%
  mutate(
    condition_standardized = case_when(
      str_detect(condition_at_discharge, "\\buhelbredet\\b") ~ "uhelbredt",
      str_detect(condition_at_discharge, "\\bhelbredet\\b") ~ "helbredt",
      str_detect(condition_at_discharge, "\\bbedret\\b") ~ "bedret",
      str_detect(condition_at_discharge, "\\bdød\\b") ~ "død",
      str_detect(condition_at_discharge, "\\bpaa prøve\\b") ~ "på prøve",
      TRUE ~ condition_at_discharge
    ),

    condition_standardized = recode(
      condition_standardized,
      !!!condition_recode,
      .default = condition_standardized
      #(everything else should be treated as NA)
    )
  )

clean_condition_counts <- df %>%
  count(condition_standardized, sort = TRUE)

cat(
  "Number of cleaned conditions:",
  nrow(clean_condition_counts))

```

```
## Number of cleaned conditions: 46
```

```
clean_condition_counts
```

```
## # A tibble: 46 × 2
##   condition_standardized    n
##   <chr>                  <int>
## 1 helbredt              1086
## 2 død                   965
## 3 bedret                780
## 4 uhelbredt            723
## 5 <NA>                 265
## 6 ikke sindssyg        70
## 7 på prøve             36
## 8 uhelb                 7
## 9 uhelbredelig         6
## 10 s                   2
## # i 36 more rows
```

```
# Retaining only defined discharge-condition categories
```

```
valid_conditions <- c(
  "helbredt",
  "bedret",
  "uhelbredt",
  "død",
  "på prøve",
  "genindlagt",
  "ikke sindssyg"
)

df <- df %>%
  mutate(
    condition_standardized = if_else(
      condition_standardized %in% valid_conditions,
      condition_standardized,
      NA_character_
    )
  )

#checking
clean_condition_counts <- df %>%
  count(condition_standardized, sort = TRUE)
clean_condition_counts
```

```
## # A tibble: 7 × 2
##   condition_standardized    n
##   <chr>                  <int>
## 1 helbredt              1086
## 2 død                   965
## 3 bedret                780
## 4 uhelbredt            723
## 5 <NA>                 317
## 6 ikke sindssyg        70
## 7 på prøve             36
```



```
# Comparing discharge conditions across the two archival series
condition_by_series <- df %>%
  filter(!is.na(condition_standardized)) %>%
  count(
    sex_series,
    condition_standardized
  ) %>%
  group_by(sex_series) %>%
  mutate(
    proportion = n / sum(n)
  ) %>%
  arrange(sex_series, desc(n)) %>%
  ungroup()

condition_by_series
```

```
## # A tibble: 12 × 4
##   sex_series condition_standardized      n proportion
##   <fct>      <chr>                <int>      <dbl>
## 1 men        død                    540      0.295
## 2 men        helbredt                 491      0.268
## 3 men        uhelbredt                 365      0.199
## 4 men        bedret                   359      0.196
## 5 men        ikke sindssyg              59       0.0322
## 6 men        på prøve                   19       0.0104
## 7 women      helbredt                   595      0.326
## 8 women      død                      425      0.233
## 9 women      bedret                   421      0.230
## 10 women     uhelbredt                 358      0.196
## 11 women     på prøve                   17       0.00930
## 12 women     ikke sindssyg              11       0.00602
```

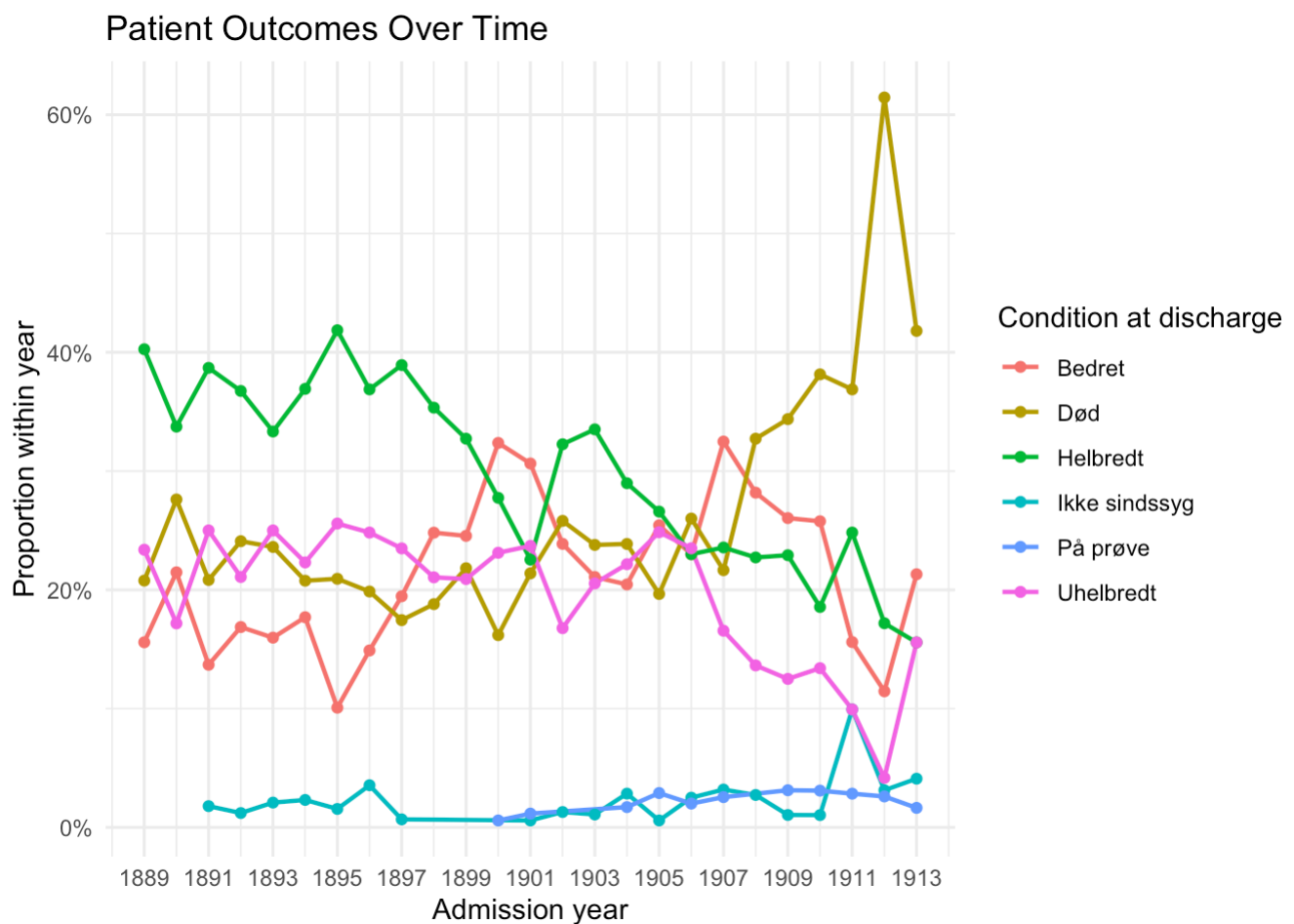
```
condition_over_time <- df %>%
  filter(
    !is.na(condition_standardized),
    !is.na(admission_year)
  ) %>%
  group_by(admission_year) %>%
  mutate(total_conditions = n()) %>%
  ungroup() %>%
  count(
    admission_year,
    condition_standardized,
    total_conditions
  ) %>%
  mutate(
    proportion = n / total_conditions
  )
```

```

condition_over_time_plot <- ggplot(
  condition_over_time,
  aes(
    x = admission_year,
    y = proportion,
    color = condition_standardized
  )
) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_x_continuous(
    breaks = seq(1889, 1913, by = 2)
  ) +
  scale_y_continuous(labels = scales::percent) +
  scale_color_discrete(
    labels = \(x) str_to_sentence(x)
  ) +
  labs(
    title = "Patient Outcomes Over Time",
    x = "Admission year",
    y = "Proportion within year",
    color = "Condition at discharge"
  ) +
  theme_minimal()

```

condition_over_time_plot



```

condition_over_time_sex <- df %>%
  filter(
    !is.na(condition_standardized),
    !is.na(admission_year)
  ) %>%
  group_by(sex_series, admission_year) %>%
  mutate(total_conditions = n()) %>%
  ungroup() %>%
  count(
    sex_series,
    admission_year,
    condition_standardized,
    total_conditions
  ) %>%
  mutate(
    proportion = n / total_conditions
  )

condition_over_time_sex

```

```

## # A tibble: 247 × 6
##   sex_series admission_year condition_standardized total_conditions     n
##   <fct>          <dbl> <chr>                                <int> <int>
## 1 men           1889 bedret                                50     6
## 2 men           1889 død                                50    13
## 3 men           1889 helbredt                             50    20
## 4 men           1889 uhelbredt                             50    11
## 5 men           1890 bedret                                89    19
## 6 men           1890 død                                89    24
## 7 men           1890 helbredt                             89    33
## 8 men           1890 uhelbredt                             89    13
## 9 men           1891 bedret                                80     7
## 10 men          1891 død                                80    23
## # i 237 more rows
## # i 1 more variable: proportion <dbl>

```

INFERENCE ANALYSES

```
# Contingency table: sex × condition at discharge
condition_sex_table <- df %>%
  filter(
    !is.na(sex_series),
    !is.na(condition_standardized)
  ) %>%
  count(
    sex_series,
    condition_standardized
  ) %>%
  pivot_wider(
    names_from = condition_standardized,
    values_from = n,
    values_fill = 0
  )

condition_sex_table
```

```
## # A tibble: 2 × 7
##   sex_series bedret   død helbredt `ikke sindssyg` `på prøve` uhelbredt
##   <fct>      <int> <int>   <int>         <int>      <int>      <int>
## 1 men          359   540     491           59        19       365
## 2 women        421   425     595          11        17       358
```

```
condition_sex_matrix <- df %>%
  filter(
    !is.na(sex_series),
    !is.na(condition_standardized)
  ) %>%
  with(
    table(sex_series, condition_standardized)
  )

condition_sex_matrix
```

```
##           condition_standardized
## sex_series bedret død helbredt ikke sindssyg på prøve uhelbredt
##      men    359 540     491           59      19       365
##      women  421 425     595          11      17       358
```

```
condition_sex_chisq <- chisq.test(condition_sex_matrix)
condition_sex_chisq
```

```
##
## Pearson's Chi-squared test
##
## data:  condition_sex_matrix
## X-squared = 61.676, df = 5, p-value = 5.473e-12
```

```
condition_sex_chisq$expected
```

```
##           condition_standardized
## sex_series  bedret      død helbredt ikke sindssyg på prøve uhelbredt
##      men   390.6393 483.291 543.8902    35.05738 18.02951 362.0926
##      women 389.3607 481.709 542.1098    34.94262 17.97049 360.9074
```

```
sum(condition_sex_chisq$expected < 5)
```

```
## [1] 0
```

```
mean(condition_sex_chisq$expected < 5)
```

```
## [1] 0
```

```
condition_sex_chisq$stdres
```

```
##           condition_standardized
## sex_series  bedret      død  helbredt ikke sindssyg  på prøve uhelbredt
##      men   -2.5542009  4.2548102 -3.8276027    5.7789190  0.3251005  0.2414078
##      women  2.5542009 -4.2548102  3.8276027   -5.7789190 -0.3251005 -0.2414078
```

```
n <- sum(condition_sex_matrix)

cramers_v <- sqrt(
  as.numeric(condition_sex_chisq$statistic) /
  (n * min(
    nrow(condition_sex_matrix) - 1,
    ncol(condition_sex_matrix) - 1
  ))
)

cramers_v
```

```
## [1] 0.1298127
```